

Descriptive Statistics and Introduction to Probability
Problem sheet N°2

Exercise 1. The following tables represent the number of order for a certain product each week during one year and the difference in scores between the winning and losing teams in a sample of 15 college football basket-ball games from 2021-2022.

No. of order	f	Point Difference	Number of Games
10 – 12	4	1 - 5	8
13 – 15	12	6 - 10	0
16 – 18	22	11 - 15	2
19 – 21	14	16 - 20	3
Total	52	21 - 25	1
		26 - 30	0
		31 - 35	1

1. Compute the characteristics of the two sets of data.
2. Compute their absolute deviation.
3. Suppose that the number of order doubled the next year with no changes in its frequencies, what would be the average and the standard deviation for this year.

Exercise 2. Given the following grades on a test:

86, 92, 100, 93, 89, 95, 79, 98, 68, 62, 71, 75, 88, 86, 93, 81, 100, 86, 96, 52

what percentage of scores lie within one standard deviation from the mean? two standard deviations? Are these results consistent with what Chebyshev's Theorem concludes?

Exercise 3. Random samples, each of size $n = 10$, were taken of the lengths in centimeters of three kinds of commercial fish, with the following results:

Sample 1	108	100	99	125	87	105	107	105	119	118
Sample 2	133	140	152	142	137	145	160	138	139	138
Sample 3	82	60	83	82	82	74	79	82	80	80

1. Draw the boxplot for each sample. Do you think their distributions are symmetric. compute pearson's coefficient of skewness to confirm.
2. which kind of fishes has the lowest variation in length.

Exercise 4. the following table shows the different values obtained for two quantitative variables X and Y in a sample of size 8

X	1.3	1.5	2.5	2.5	2.7	3	4	5
Y	2.3	3.5	3.5	4.5	4.7	3	2	1

1. Draw the scatter plot.
2. Calculate the covariance and the linear correlation coefficient.
3. Can we conclude in this case that there is a linear link between X and Y? What does the scatter plot suggest instead?

Exercise 5. A survey is carried out with a clientele (533 individuals) to study their willingness to pay (X in DA) a telephone plan. This clientele was segmented according to a criterion Y which takes 3 modalities A, B and C. After consultation, we obtain the following contingency table

$X \setminus Y$	A	B	C	Total
]0-5]	38	11	0	49
]5-10]	55	63	0	118
]10-15]	53	76	0	129
]15-20]	32	62	1	95
]20-25]	6	24	8	38
]25-30]	5	4	40	49
]30-35]	2	1	37	40
]35-40]	1	0	12	13
]40-45]	0	0	2	2
Total	192	241	100	533

We give the following partial results
 $\sum_i n_{iA}x_{iA} = 2105$, $\sum_i n_{iA}x_{iA}^2 = 31750$
 $\sum_i n_{iB}x_{iB} = 3217.5$, $\sum_i n_{iB}x_{iB}^2 = 50706.25$
 $\sum_i n_{iC}x_{iC} = 3035$, $\sum_i n_{iC}x_{iC}^2 = 94175$

1. Determine the conditional means of X. Deduce the marginal mean of X.
2. Determine the marginal and conditional variances.
3. Compute the correlation between the three modalities of Y two by two. Do you think that this segmentation makes sense?

Exercise 6. The following table contains the information of age, monthly income, health condition and frequency of exercise last month of 10 respondents.

Age (years)	Monthly Income (DA)	Health Condition	Frequency of exercise per month
62	68080	Very good	22
36	48000	Very good	22
55	84100	Good	15
55	77000	Fair	16
27	48510	Excellent	30
37	38400	Poor	5
49	68560	Poor	10
49	56380	Poor	10
26	22970	Good	22
38	44330	Fair	13

1. Calculate the correlation coefficient between the age and the monthly income. Hence, interpret the value obtained.
2. Draw a scatter diagram to represent the relationship between the age and the monthly income and show the point $(\bar{X}, \bar{Y})$. Interpret your result.
3. Draw the cross tabulation then calculate the Spearman's rank correlation coefficient between health condition and frequency of exercise per month. Hence, comment on the result obtained